# Hongpei Li

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## Education

Northwestern University — 2026 - 2031 (Expected)
Ph.D. in Industrial Engineering and Management Sciences — Evanston, IL, USA
Advisor: Zikai Xiong

Shanghai University of Finance and Economics — 2021 - 2025
Bachelor of Engineering, Pilot Class of Interdisciplinary Sciences — Shanghai, China

## Research Interests

Large-scale optimization on AI-era computing hardware; intersection of AI and optimization.

## Publications

Hongpei Li, Han Zhang, Ziyan He, Jiaqi Hu, and Hongsong Yuan. A Learning and Searching Method for Integrated Process Planning and Scheduling.
*Production and Operations Management*, 2026. doi: 10.1177/10591478261505544. Code: github.com/Lhongpei/ipps-drl.

Hongpei Li, Hui Yuan, Han Zhang, Jianghao Lin, Dongdong Ge, Mengdi Wang, and Yinyu Ye. FMIP: Joint Continuous-Integer Flow for Mixed-Integer Linear Programming.
*ICLR 2026*. arXiv:2507.23390. Code: github.com/Lhongpei/FMIP.

Yicheng Huang, Wanyu Zhang, Hongpei Li, Dongdong Ge, Huikang Liu, and Yinyu Ye. A Restarted Primal-Dual Hybrid Conjugate Gradient Method for Large-Scale Quadratic Programming.
*INFORMS Journal on Computing*, 2025. doi: 10.1287/ijoc.2024.0983.

## Preprints

Hongpei Li, Yicheng Huang, Huikang Liu, Dongdong Ge, and Yinyu Ye. GPU-Accelerated Conic Quadratic Programming with Local Linear Convergence under Strict Complementarity.
2026. Under review at *Mathematics of Operations Research*. arXiv:2608.09159.

Hongpei Li, Huikang Liu, Dongdong Ge, and Yinyu Ye. A Curvature-Aware Rank-Adaptive Distributed Augmented-Lagrangian Solver for Large-Scale SDPs.
2026. arXiv:2607.17933.

Chenyu Zhou, Hongpei Li, Yuerou Liu, Jianghao Lin, Dongdong Ge, and Yinyu Ye. OSDN: Improving Delta Rule with Provable Online Preconditioning in Linear Attention.
2026. arXiv:2605.13473.

Hongpei Li, Yicheng Huang, Huikang Liu, Dongdong Ge, and Yinyu Ye. PDHCG-II: An Enhanced Version of PDHCG for Large-Scale Convex QP.
2026. arXiv:2602.23967.

## Preprints (continued)

Hongpei Li, Yicheng Huang, Huikang Liu, Dongdong Ge, and Yinyu Ye. D-PDLP: Scaling PDLP to Distributed Multi-GPU Systems.
2026. Under review at *ICLR 2027*. arXiv:2601.07628.

Hongpei Li, Han Zhang, Huikang Liu, Dongdong Ge, and Yinyu Ye. OptPipe: Memory- and Scheduling-Optimized Pipeline Parallelism for LLM Training.
2025. arXiv:2510.05186.

Huikang Liu, Yicheng Huang, Hongpei Li, Dongdong Ge, and Yinyu Ye. A Scalable First-Order Method for Large-Scale Competitive Market Equilibrium Computation.
2025. Under review at *INFORMS Journal on Computing*. arXiv:2506.06258. Code: github.com/Lhongpei/PDNLP.

Hongpei Li, Ziyan He, Yufei Wang, Wenting Tu, Shanwen Pu, Qi Deng, and Dongdong Ge. BenLOC: A Benchmark for Learning to Configure MIP Optimizers.
2025. arXiv:2506.02752. Code: github.com/Lhongpei/BenLOC.

## Software

- PDHCG: GPU-accelerated primal-dual hybrid conjugate gradient solvers for quadratic and conic quadratic programming. github.com/Lhongpei/PDHCG
- CARDAL: A multi-GPU, rank-adaptive augmented Lagrangian solver for large-scale semidefinite programming. github.com/Lhongpei/CARDAL
- D-PDLP: A distributed primal-dual first-order solver for large-scale linear programming across multiple GPUs. github.com/Lhongpei/D-PDLP

## Work Experience

QingTian Lab, Cardinal Operations — Feb. 2025 - Jun. 2026
LLM & OPT Researcher
Developed GPU-accelerated optimization solvers for COPT and conducted research on large language model inference.

## Teaching Experience

Shanghai University of Finance and Economics
- Advanced Operations Research
  Teaching Assistant, Spring 2024
- Object-Oriented Analysis & Design
  Teaching Assistant, Fall 2023
- Programming Peer Tutor
  Fall 2022

## Honors & Awards

- Honor Bachelor Degree, 2025 (top 5% in major)
- Honor of Excellence Graduation Thesis, 2025 (top 3% in major)
- Shanghai Municipal Bronze Award, Internet+ University Student Innovation and Entrepreneurship Competition (0.13%)